1. Grey-Level Transformations for a Dark Image.

Grey-level transformations such as log transformation and power-law (gamma) transformation can be used to improve the brightness of image.

1) log transformation:

The formula is:

$$S = c \log(1+r)$$

where r is the input pixel intensity, S is the output intensity, and c is a constant.

* It expands low-intensity pixel values.
* It makes details in dark regions more visible.
* It is useful when the img. contains many dark pixels.

2) Power-law (gamma) transformation:

The formula is:

$$S = cr^{\gamma}$$

* when $\gamma < 1$ the image becomes brighter.
* When $\gamma > 1$ the image becomes darker.
* A gamma value below 1 is suitable for this problem.

processing steps:

1. Read the original image
2. Normalize pixel values to the range 0-1

3. Apply log or gamma transformations.

4. Rescale the output intensity values.

5. Display the enhanced image and compare it with the original.

2. Histogram Equalization for fingerprint Enhancement:

Histogram Equalization is an image enhancement technique that improves contrast by redistributing pixel intensity values over a wider range.

WORKING PRINCIPLE:

* A histogram represents the frequency of each intensity level in an image.

* When pixel values are concentrated in a narrow range, the fingerprint has low constrast.

* Histogram equalization spreads the intensity values across a wider range.

* This improves the visibility of fingerprint ridges and valleys.

Processing Steps:

1. Read the fingerprint image.

2. Convert it grayscale if neccessary.

3. Calculate the image histogram.

4. Compute the cumulative distribution function (CDF)

5. Map the original intensity values to new values using the CDF.

6. Display the enhanced fingerprint.

PROCESSING STEPS:

1. Read the fingerprint image.

2. Convert it into grayscale if necessary.

3. Calculate the img. histogram.

4. Compute the cumulative distribution function (CDF)

5. Map the original intensity values to new values using the CDF.

6. Display the enhanced fingerprint.

The transformation is commonly expressed as:

$$S_k = (L-1) \sum_{j=0}^{k} p(r_j)$$

Here, L is the number of intensity levels and $p(r_j)$ is the probability of intensity r_j.

3) Salt- and pepper noise removal

The median filter is suitable for removing salt and pepper noise while preserving edges.

* Select a 3×3 neighborhood around each pixel.
* Arrange the pixel values in ascending order.
* Replace the central pixel with the median value.
* Repeat the process for the entire image.

Why not an averaging filter? An averaging filter is affected by extreme noise values and may blur edges, whereas a median filter handles isolated black and white pixels more effectively

4. Laplacian Sharpening and unsharp masking

1. Laplacian sharpening :-

Highlights rapid changes in img intensity

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

It enhances edges and makes copper tracks more visible.

2. Unsharp masking:

* Blur the original img using smoothing filter
* Subtract the blurred img. from original to create a detail mask
* Add the detail mask to orginal img.

$$g = f + k(f - f_{blurred})$$

Here, k controls the sharpening strength.

5. Frequency - Domain Low - Pass Filtering:

A low-pass filter removes high-frequency components to reduce noise and smooth the image. Common filters include ideal, Butterworth and Gaussian low-pass filters.

Processing steps:

1. Convert the img. to grayscale if necessary.
2. Apply the Fourier transform.
3. Multiply the frequency spectrum by a low-pass filter mask.
4. Apply the inverse Fourier transform.
5. Display the filtered image.

Effect of cutoff frequency:

* Low cutoff
* High cutoff

6. High-Pass Filtering:

A high pass filter enhances high-frequency components including edges and fine details.

Processing steps:

and.

1. Read the retinal image and select a suitable grayscale
2. Suppress low-frequency components using a high-pass filter.
3. Display the enhanced image.

High boost filtering can also be used to preserve the original image.

$$g = Af - f_{blurred} \quad , A > 1$$

Point Detection and Line Detection Masks

Spatial masks detect image features by examining intensity differences between neighbouring pixels.

1. Point detection:

Use the following mask to detect isolated points:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Apply the mask and compare the response with a threshold. Strong responses indicate possible air bubbles.

2. Line detection:

Line masks identify horizontal, vertical or diagonal cracks. For ex, a horizontal line mask is:

$$\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Different masks can detect cracks in different directions.

Threshold selection :-

* A low threshold detects more features but may produce false detection.

* A high threshold reduces false detections but may miss faint defects.

8) Thresholding for Handwritten Answer Sheets

Adaptive thresholding is suitable because it calculates thresholds for local regions rather than using one threshold for the entire regions.

Global thresholding: Uses a single threshold T for all pixels.

$$g(x,y) = \begin{cases} 1, & f(x,y) > T \\ 0, & f(x,y) \leq T \end{cases}$$

9) Region Growing

Region growing is an image segmentation technique that groups neighbouring pixels and add those satisfying the threshold.

The similarity condition can be written as:

$$|I(x,y) - I_{seed}| < T$$

T is similarity threshold

Seed selection: Choose a pixel inside the lake with representative intensity.

10) Edge Linking and Boundary Detection:-

Edge linking connects broken edge segments to obtain a complete object boundary.

1. Local edge linking:-

Detects edges using Sobel or Canny, Examine neighboring edge pixels, Apply morphological Closing.

2. Hough transform:

$$\rho = x\cos\theta + y\sin\theta$$

It transform detects geometric shapes by mapping edge pixels into parameter space.